

Case Study 3: How Can an E-Commerce Company Increase Customer Engagement and Repeat Purchases?

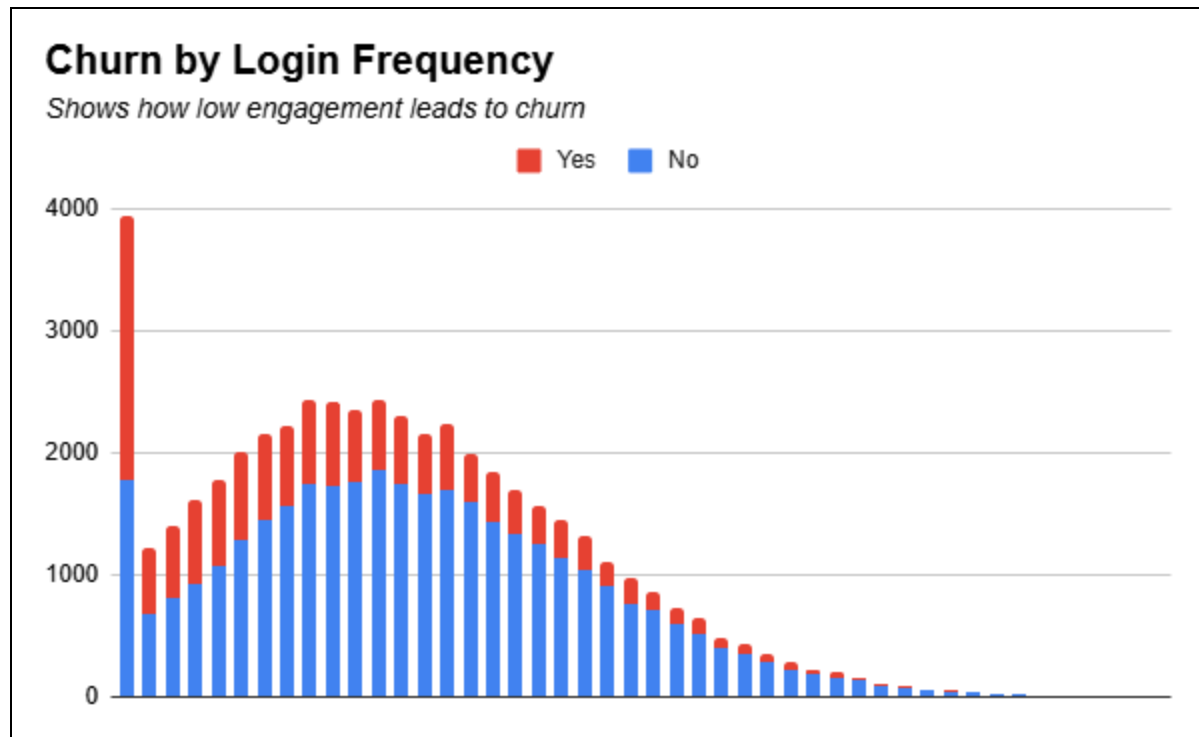
Intro: In this case study, I explore how an e-commerce success depends on keeping customers engaged and encouraging repeat purchases. This case study explores how NovaMart, a mid-sized online retailer, can use customer behavior data to understand what drives engagement and loyalty. By analyzing patterns in login frequency, session duration, and mobile app usage, the goal is to uncover insights that help NovaMart strengthen relationships and boost sales.

Problem:

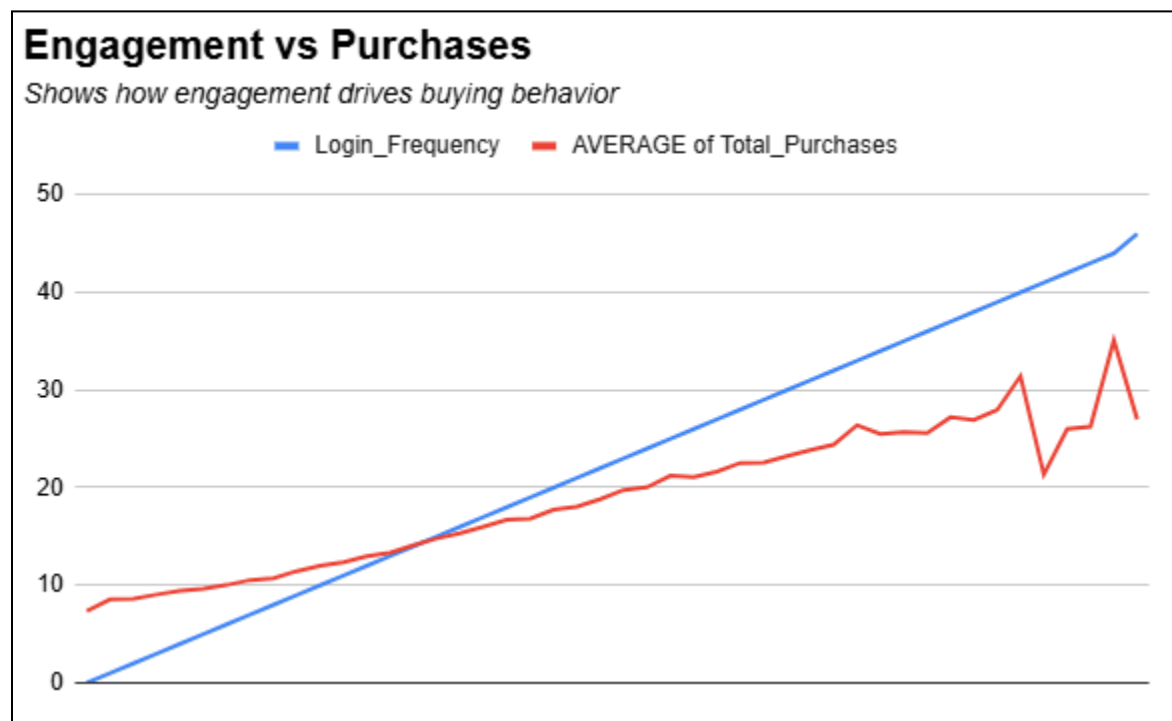
1. Which behaviors (login frequency, session duration, mobile app usage, social media engagement) are most strongly linked to repeat purchases?
2. How do engagement metrics relate to total purchases and average order value?
3. How can NovaMart use engagement data to predict and reduce churn?

NovaMart wants to understand how customer engagement influences repeat purchases and retention. The company collects data on behaviors such as login frequency, session duration, mobile app usage, and social media engagement, but it lacks clarity on how these activities translate into sales and loyalty. This analysis aims to uncover which engagement behaviors are most strongly linked to repeat purchases, how they relate to total purchases and average order value, and how NovaMart can use these insights to predict and reduce churn.

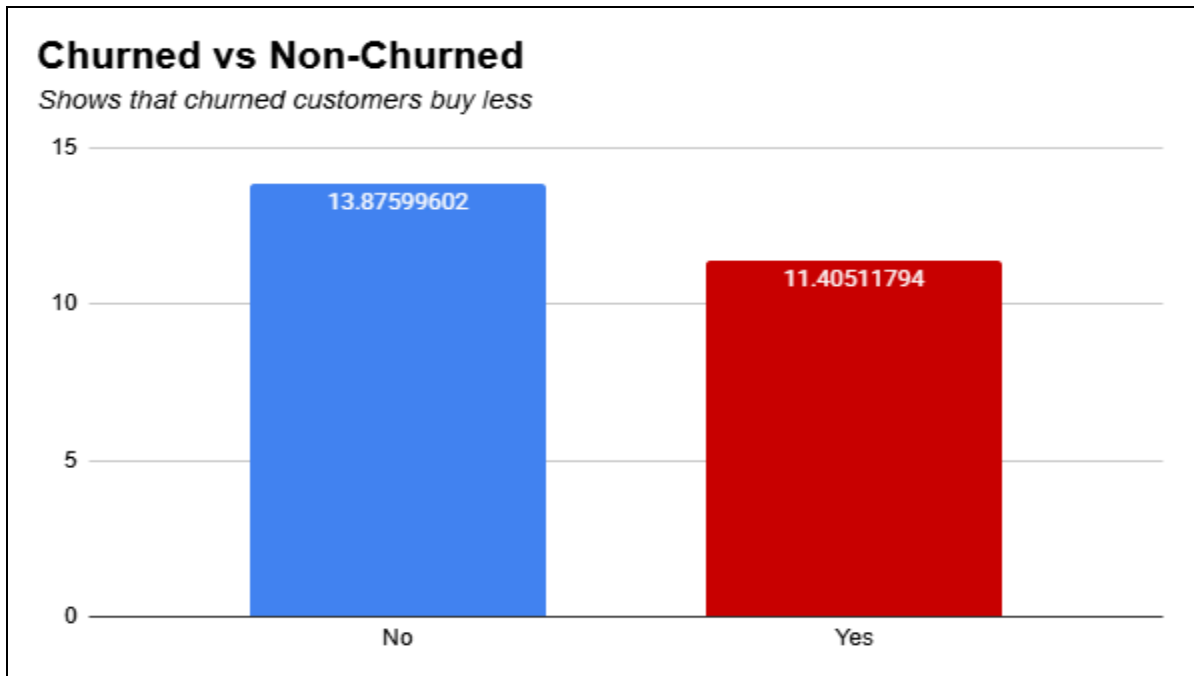
Summary: For this case study, I analyzed NovaMart's customer behavior data using Google Sheets to uncover patterns in engagement and repeat purchases. Through pivot tables and charts, I examined relationships between login frequency, mobile app usage, email engagement, and purchase activity. The analysis revealed that customers with higher engagement, which were measured by frequent logins, app usage, and email interaction, tend to make more purchases and remain as loyal customers. Inactive and lower active customers are at the highest risk of churn.



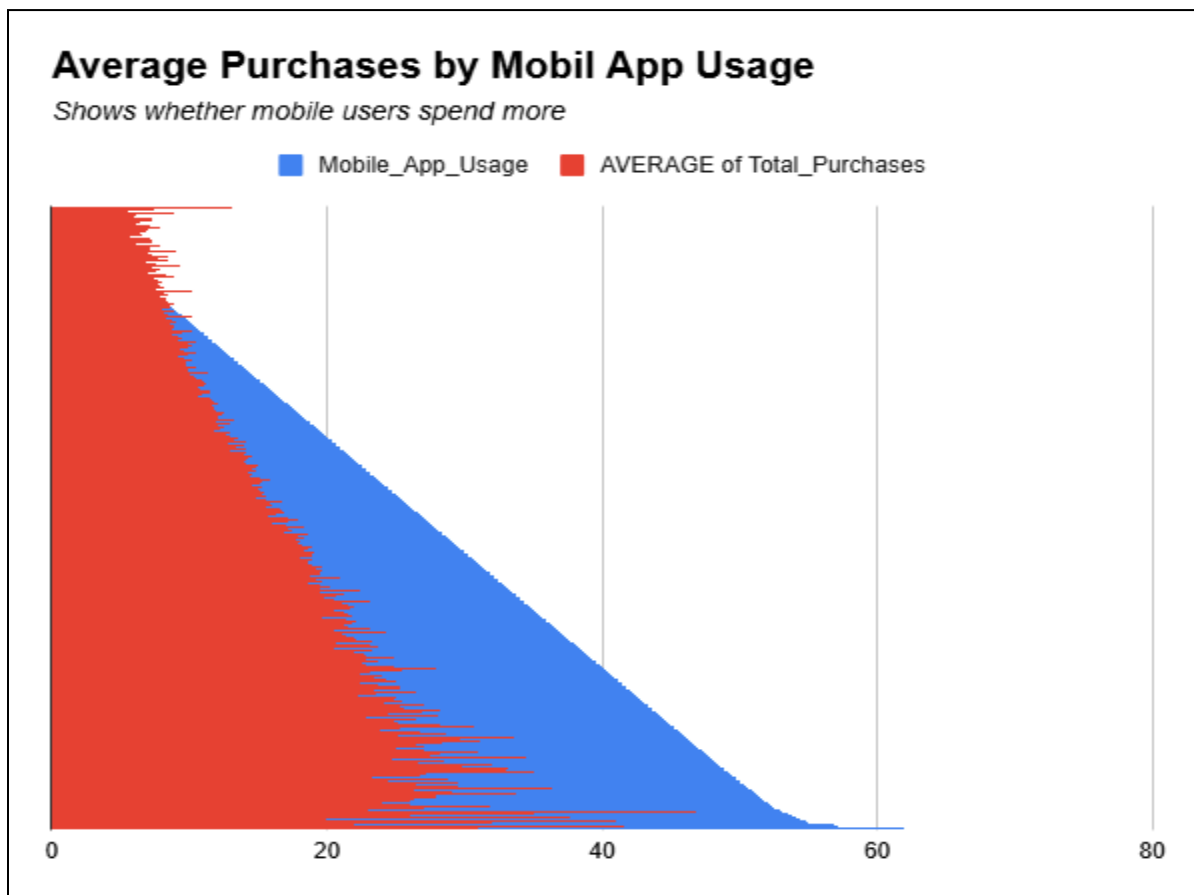
Insight: Customers with 0-1 logins have the highest churn rates, showing that inactivity strongly predicts churn.



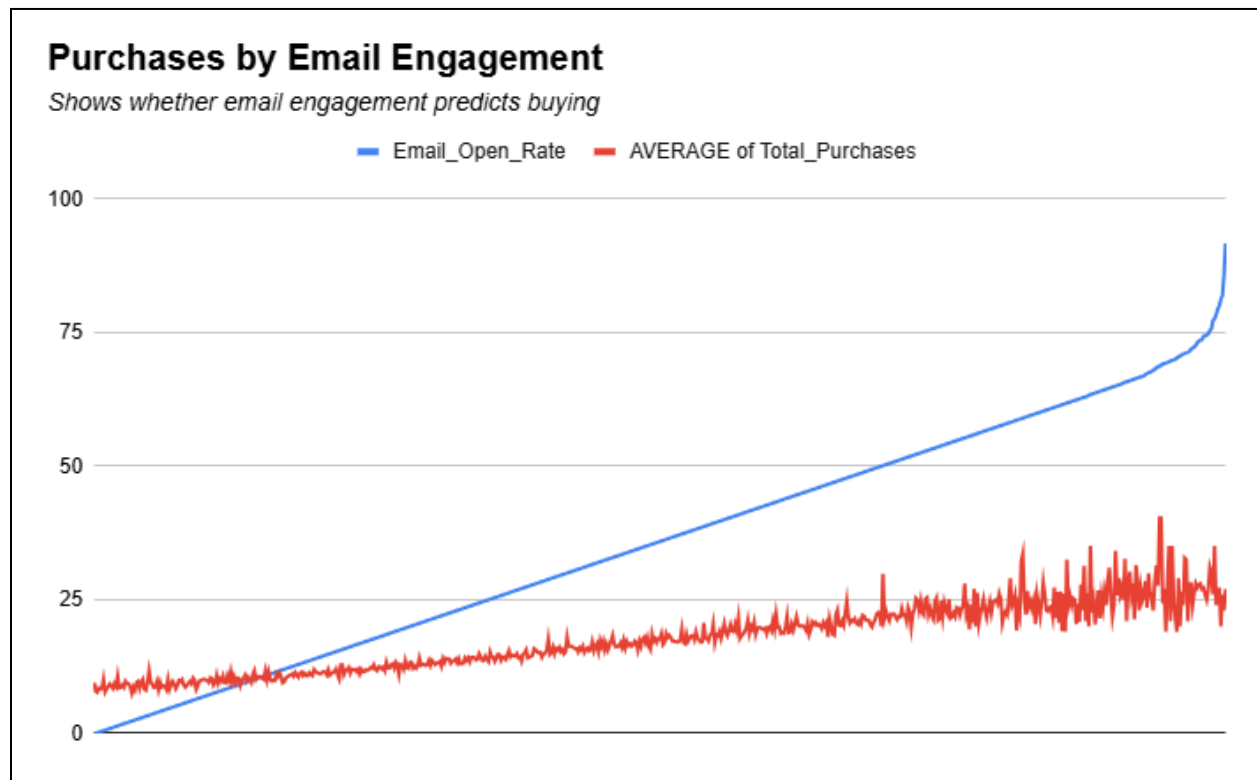
Insight: Average purchases increase steadily with higher login frequency, indicating that engaged customers spend more.



Insight: Customers who churn tend to have significantly fewer purchases than those who stay.



Insight: Mobile app users show higher average purchases than non-app users, showing that app users are more likely to purchase.



Insight: Customers with higher email rates tend to make more purchases, email engagement drives sales.

Key Findings:

- Low login frequency strongly correlates with high churn.
- Customers who engage more frequently make more purchases and show higher loyalty.
- Mobile app users spend more and purchase more often than non-app users.
- Higher email open rates are linked to increased purchase activity.
- Purchases behavior is a strong indicator of customer retention , churned users buy less overall.

Recommendations:

- Re-engage with inactive users, send personalized reminders, discounts, or loyalty rewards to customers with low login frequency.
- Enhance mobile app experience. Promote app downloads and offer exclusive app only deals to increase engagement.
- Optimize email marketing. Use Personalized subject lines, targeted product suggestions, and ideal send times to boost open rates.

- Reward repeat purchases. Introduce a tier loyalty program for frequent buyers to strengthen retention.
- Monitor low-purchase customers. Identify users with declining purchase activity early and target them with retention campaigns.

Conclusion: Through this analysis, I was able to explore NovaMart's customer behavior data in depth by using Google Sheets. Creating pivot tables and visual charts, I identified clear patterns showing how engagement metrics, such as login frequency, mobile app usage, and email interaction, directly influence repeat purchases and customer retention. This hands on strengthen my data-analysis skills while uncovering actionable insights that NovaMart can use to reduce churn and improve loyalty. Overall, this project demonstrated how data-driven decisions can guide effective business strategies and enhance customer relationships.